Linear CRT radii and semantic radix recovery

An explicit quartic for syntactic ZFC consistency

AI-assisted mathematical research

Prepared for Vladimir Reshetnikov and the ProveIt project

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Abstract

We give a completely specified quartic in 238 unrestricted integer variables whose unsolvability is provably equivalent in ZFC to syntactic consistency of ZFC, using the identified inherited numerical-calculus calibration. The equation is the sum of squares of 174 printed quadratic residuals; its collected expansion has 13,149 monomials and coefficient height 6,144. Two 239-variable companions trade support against height.

The arithmetic reduction uses a single growth value $F = t^t$ for both proof-label separation and a binomial CRT radius. An even index chart and an affine majorant that cannot be a unit recover the radix without a private square witness. An existing power request then recovers the sign of the beta spacing, permitting its odd multiplier to be unsquared. Three arithmetic coordinates disappear, and exact reversible row operations give the reported polynomials. Complete coefficient files, constructive proofs, restoration certificates, independent finite checkers, and deterministic rebuild code accompany the article.

This is an explicit improvement in the inspected direct small-coefficient quartic lineage. It is neither a proof of consistency, a global minimality claim, nor a newly compiled end-to-end proof-assistant formalization.

Comparison snapshot:
`7a8da414376a80b90ecf799f37ceb26fb9ac674e`
All unknowns in the three new equations range over $\mathbb{Z}$.

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1 Result and comparison

The principal object of this report is the fixed ordinary polynomial

$$\boxed{P_{238}(x_0,\dots,x_{237}) = \sum_{i=0}^{173} R_i(x_0,\dots,x_{237})^2 = 0, \qquad (x_0,\dots,x_{237}) \in \mathbb{Z}^{238}.} \tag{1}$$

All 174 quadratic residuals are printed in Appendix A. The accompanying file `equations/P238/equation.json` gives the same complete list, and `expanded_polynomial.json` in that directory gives every collected coefficient. Thus the sum in (1) specifies one polynomial with fixed integer coefficients; it is not a variable-length system or a universal-polynomial placeholder.

Theorem 1.1 (Consistency equivalence)**.** *For the explicitly specified polynomial* (1)*, the ordinary arithmetic proof below, composed with the identified inherited numerical calculus calibration, establishes*

$$\mathrm{ZFC} \vdash \Big(\neg\exists \boldsymbol{x} \in \mathbb{Z}^{238}\ P_{238}(\boldsymbol{x}) = 0\Big) \longleftrightarrow \mathrm{Con(ZFC)}.$$

The same assertion holds for the two accompanying 239-variable polynomials, with their specified domains and coefficients.

Here consistency is syntactic consistency: no finite proof of falsity from the fixed ZFC axiom predicate. The proof assumes neither consistency nor inconsistency. It supplies an ordinary mathematical argument formalizable in ZFC; it does not supply a newly compiled end-to-end Lean/Rocq theorem or an encoded object-language ZFC proof string.

1.1 Exactly what improved

Equation over $\mathbb{Z}$	Variables	Rows	Monomials	Height
Incoming Shared Factorial Error	240	177	21,482	6,144
Incoming factorial-free, sparse	242	178	5,018	7,968
Incoming factorial-free, dense	241	177	12,774	7,968
New P_{238}	**238**	**174**	**13,149**	6,144
New $P_{239}^{\mathrm{sparse}}$	239	175	**4,963**	7,968
New P_{239}^{low}	239	175	6,723	**4,432**

Every polynomial in the table has exact degree four. “Rows” means the number of residuals whose squares are summed; “monomials” counts terms after complete collection; height is the largest absolute collected coefficient. Every coefficient of an individual new residual has absolute value at most 64. Every declared variable occurs in the collected polynomial. The comparison equations are retained and recollected by the verifier.

Against the smallest-dimension direct quartic in the seven incoming archives at the pinned snapshot, P_{238} removes two variables, three rows, and 8,333 monomials, with unchanged height. The new low-height 239-variable equation improves all four size measures of that 240-variable comparison. Against its immediate factorial-free sparse predecessor, $P_{239}^{\mathrm{sparse}}$ removes three variables, three rows and 55 monomials, with unchanged height.

These are improvements to the direct numerical-calculus construction. The repository also has much smaller variable counts in other coefficient and degree architectures, including a 37-integer quartic and high-degree few-variable forms [1]. The present result does not establish a global simplicity minimum. Its advantages are a fully specified quartic, small literal coefficients, and a direct proof-calculus interpretation.

1.2 The mathematical contribution

The immediate parent has already removed factorial authenticity from its CRT construction [2]. This report removes three more arithmetic coordinates by a coordinated argument:

1. The single growth value $F = t^t$ suffices both to distinguish bounded proof-label products and to serve as the CRT radius. Thus $R = F$ replaces $R = F^2$.
2. An even CRT index and a universally nonunit affine majorant make the power base nonzero before any representative sign is known. This removes the separate square restriction on the radix.
3. An existing power request recovers the sign of the beta spacing. Its odd multiplier can therefore be affine, instead of squared.

The elementary LCM bound and polynomial coefficient arguments are proved here, without a priority claim for those general facts. The new contribution is their use together with a noncircular sign-recovery order in this explicit source. The final row-basis reductions are reversible integer linear algebra, not additional logical assumptions.

2 Construction and inherited interfaces

The source uses the fixed numerical proof calculus C_{222}, with truth code 0 and falsity code 1. A finite multiset of locally valid, single-conclusion rules is certified arithmetically. Beta coding stores the finite accumulator sequence; a binomial CRT certificate checks all local rows at once; a shared Pell construction certifies the finite set of required powers. This route uses neither a simulation of K215 nor a specialization of Jones's universal equation. General Diophantine representation results provide historical context, rather than an unspecified coefficient in this construction [5].

The unchanged source supplies fourteen principal Pell pairs, eighteen power requests, a signed frame, a semantic anchor, and one common norm bound for eleven modular remainders. The following interfaces are retained with their ordinary proofs and source identities:

Interface	Consequence used here
Signed Pell frame and anchor	Canonical signed-index Pell coordinates, before arithmetic signs; every actual base, index and output has absolute value below W, where $W \geq 258$ and $\lvert A\rvert > W^W$.
Local constructor ledger	Derived-label bound $h_* \leq H^{2^{22}}$ after representatives are nonnegative; local nonnegative row value below $(2 + t + x + y + F)^{2^{23}}$.
Horn calibration	Valid local rules and the recovered label balance imply a finite C_{222} derivation of code 1.
Logical calibration	C_{222} refutability is equivalent to syntactic ZF refutability, and then to syntactic ZFC refutability.

The named logical interfaces are

`KEndpoint.c222_iff_zf_refutation`,
`ConstructibleChoice.zf_refutation_iff_zfc_refutation`.

Their source files are included under the inherited calibration directory. The first theorem is about the numerical calculus itself; the machine simulation hypotheses in other theorems of its source file are not premises of this theorem. The second interface uses the constructible-universe interpretation to transport ZFC refutations to ZF refutations. This report does not rebuild their complete transitive proof-assistant environment.

3 The new arithmetic choices

Use unrestricted integer certificate coordinates and the semantic values

$$\begin{gathered} t = 16 + 4h^2, \qquad F = t^t, \qquad R = F, \qquad C = 2c, \\ H = 2 + 4\left(C + \sum_{j=1}^{8} A_j\right), \qquad y = F(2v+1). \end{gathered} \tag{2}$$

The nine actual CRT representatives are $C, A_1, \ldots, A_8$. Historical stored coordinates may be these values minus one. The literal source mutation consistently changes the actual index to $C = 2c$.

The retained endpoint and CRT equations include

$$Bh_B = C + F + 1, \tag{3}$$

$$x = F + 1 + (y+1)u_0, \tag{4}$$

$$x = 1 + [1 + (t+1)y]u_t. \tag{5}$$

The power graphs request

$$F = t^t, \qquad q = (BH)^x, \qquad N = B^x, \qquad M = q^x, \qquad D_y = q^{y+1},$$

as well as the binomial numerator and denominator powers. At the start of soundness, these displayed semantics must be recovered from their polynomial certificates. In particular, no positive value of B, q, x, y or a representative is assumed.

Variable exponentiation is explanatory notation in this section. The final residuals contain only integer constants, addition, subtraction and multiplication. The existing first power slot is reused with base and index both t; no nineteenth power request is introduced. The old affine base slot is retained with value t, so the frame counts its square twice. This harmless duplication strengthens the bound and omits no actual atom.

4 A sharp enough growth value for label separation

For a label $j \in \{0, \ldots, t\}$ set $W_j = 1 + jF$, so $W_0 = 1$. There are t local rows, each with one conclusion and at most two premises. The desired product balance is

$$\prod_{i<t} W_{v_i} = (1+F) \prod_{i<t} W_{p_i} W_{q_i}. \tag{6}$$

The symbol q_i here denotes a premise label, not the arithmetic radix q.

Lemma 4.1 (Label separation at $F = t^t$)**.** *For $t \geq 1$ and $F = t^t$, equality (6) implies that, for each positive label j, its conclusion multiplicity minus its premise multiplicity is 1 when $j = 1$ and 0 otherwise.*

Proof. Discard unit factors with label zero. If the two remaining factor counts differ, let r be their minimum. One side originally has at most t factors, so $r \leq t$. The side with more factors is at least $(F+1)^{r+1}$; the other is at most

$$(1+tF)^r \leq t^r (F+1)^r.$$

This contradicts $F + 1 = t^t + 1 > t^r$. Thus both sides have the same number $m \leq t$ of nonunit factors.

Replace F by an indeterminate z. For $m > 0$ and $0 \le k \le m$, the coefficient of z^k in either product of m factors $1 + jz$, $1 \le j \le t$, is a positive integer. It is at most

$$\binom{m}{k} t^k = \binom{m}{m-k} t^k \le m^{m-k} t^k \le t^m \le F.$$

Because both coefficients are positive, the corresponding difference has absolute value at most $t^m - 1 < F$. This strict inequality remains valid at the boundary $m = t$; using only an upper bound by F without positivity would not suffice. If $m = 0$, both products are already the polynomial 1.

A nonzero integer polynomial whose coefficients have absolute value below F cannot vanish at F: divide its evaluation by F to the degree of its lowest nonzero coefficient, then reduce modulo F. That coefficient would be a nonzero multiple of F of smaller absolute value. Therefore the evaluated equality lifts to an identity in $\mathbb{Z}[z]$. The distinct labels give distinct roots $-1/j$ of the linear factors $1 + jz$. Unique factorization over $\mathbb{Q}[z]$ gives the claimed multiplicities. □

This replaces the parent's sufficient condition $F > (t+1)^t$. That older condition is false for the new F and is not silently retained. No coprimality of the evaluated integers W_j is needed.

5 The CRT radius can be linear

Write $\Lambda_t = \operatorname{lcm}(1, \dots, t)$.

Lemma 5.1. *For every $t \ge 1$, $\Lambda_t \le 4^{t-1}$.*

Proof. Let $m = \lceil t/2 \rceil$. For each prime p, the exponent of p in Λ_t exceeds that in Λ_m by at most one. If it exceeds it, a power p^a lies above m and at most t. Its contribution to the factorial valuation of $\binom{t}{\lfloor t/2 \rfloor}$ is one; all other contributions are nonnegative. Consequently

$$\Lambda_t \mid \Lambda_m \binom{t}{\lfloor t/2 \rfloor}.$$

For $t = 2n$, use strong induction and $\binom{2n}{n} \le 4^n$ to obtain $\Lambda_{2n} \le 4^{n-1} 4^n = 4^{2n-1}$. For $t = 2n+1 > 1$, the two equal middle binomial coefficients have sum at most 2^{2n+1}, so each is at most 4^n. Hence $\Lambda_{2n+1} \le 4^n 4^n = 4^{2n}$. The base case $t = 1$ is immediate. □

It follows directly from $t \ge 16$ that

$$R = F = t^t \ge 16^t > 16^{t-1} \ge \Lambda_t^2. \tag{7}$$

Thus the existing growth value already meets the binomial-CRT radius requirement. The separate square register F^2 is unnecessary.

For clarity, the two prime-power facts used by that compiler are reproduced next. The classical binomial-LCM divisibility in their proof is also related to Farhi's binomial-row identity [4].

Lemma 5.2 (Two-sided prime-power coverage)**.** *Let $R \ge \Lambda_t^2$ and $B = \binom{R+t}{t}$. Then:*

1. *For every $i \in \{1, \dots, t\}$, some full prime power $p^e \| B$ divides $R + i$ and satisfies $p^e > t$.*
2. *Every full prime power $p^e \| B$ divides $R + i$ for at least one $i \in \{1, \dots, t\}$.*

Proof. First, $t\binom{t-1}{i-1} \mid \Lambda_t$. At a prime power p^a, the factorial valuation summand

$$\left\lfloor \frac{t}{p^a} \right\rfloor - \left\lfloor \frac{i-1}{p^a} \right\rfloor - \left\lfloor \frac{t-i}{p^a} \right\rfloor$$

is 0 or 1 and vanishes for $p^a > t$.

Fix i. Since $R + i > \Lambda_t^2$, there is a prime with $v = v_p(R+i) > 2a$, where $a = v_p(\Lambda_t)$. For $j \neq i$, $v_p(R+j) = v_p(j-i)$, because $p^v > t > |j-i|$. Therefore

$$e = v_p(B) = v - v_p\left(t\binom{t-1}{i-1}\right).$$

The subtracted quantity lies between 0 and a, so $e > a$ and $e \leq v$. Thus $p^e > t$ and $p^e \mid R+i$, proving the first assertion.

For the second assertion, set $v = \max_i v_p(R+i)$. The valuation of B is a sum of 0-or-1 factorial summands. Above level v, the interval $R+1, \ldots, R+t$ contains no multiple of that prime power; all such summands vanish. Hence $e = v_p(B) \leq v$. A row attaining the maximum has $p^e \mid R+i$. □

The second conclusion is essential for completeness. Choosing just one selector for each row would leave unspecified how to satisfy the CRT congruences at the remaining primes of B.

6 Signed power certification and the recovery order

The retained Pell interface can be used with a negative base. Let $X_k(A), Y_k(A)$ denote the usual Pell polynomials, continued to negative indices by $X_{-k} = X_k$, $Y_{-k} = -Y_k$. For a requested base $n \neq 0$, put $T_n = 2An - n^2 - 1$. The power congruence and Pell recurrence give

$$m \equiv n^k \pmod{T_n} \quad (k \geq 0), \qquad mn^{-k} \equiv 1 \pmod{T_n} \quad (k < 0).$$

The signed frame bounds the relevant absolute errors by less than $|T_n|$. For $|n| \geq 2$, a negative k would therefore force the impossible equality $mn^{-k} = 1$ in integers. A nonnegative k forces $m = n^k$. This is the precise inherited signed-power interface used below; no assertion at base zero is needed.

Proposition 6.1 (Acyclic sign recovery)**.** *Every integer zero of the changed source satisfies*

$$F > 2^{32}, \quad y \geq F > 0, \quad x \geq 2 + (t+1)y, \quad C, A_j \geq 0, \quad B > 0, \quad q > 0,$$

with the intended power and binomial meanings.

Proof. The steps must be taken in the following order.

1. Recover F and exclude a zero base. The request with known positive base and index t gives $F = t^t \geq 16^{16} = 2^{64}$. Since $t = 16 + 4h^2$, F is even. The right side of (3) is odd even if C is negative. Thus B and h_B are odd and nonzero. Independently, $H \equiv 2 \pmod 4$ for all signed representatives, so $|H| \geq 2$. The power base therefore satisfies $|BH| \geq 2$ without a positivity assumption on any representative.

2. Recover x and make the radix nonunit. The request at base BH gives $x \geq 0$ and $q = (BH)^x$. The affine beta definition gives $|y| \geq F$, because $2v + 1 \neq 0$. Hence

$$|1 + (t+1)y| \geq (t+1)F - 1 > 1.$$

If $x = 0$, the last endpoint would make this integer divide -1, which is impossible. Thus $x \geq 1$ and $Q = |q| \geq 2$.

3. Recover the beta sign from an existing power. The already present request with base q and exponent $y+1$ is now usable. It forces $y+1 \geq 0$. A negative y would satisfy $y \leq -F < -1$, so $y \geq F > 0$. This is the step that permits the unsquared odd beta multiplier.

4. Obtain the large exponent before representative signs. If $u_0 \leq -1$, (4) would give $x \leq F - y \leq 0$. Therefore $u_0 \geq 0$ and $x \geq F + 1$. Equation (5) then gives $u_t \geq 1$ and

$$x \geq 2 + (t+1)y \geq 2 + (t+1)F. \tag{8}$$

Consequently $Q \geq 2^x$ and $|q+1| \geq Q - 1 \geq 2$.

5. Recover the representative signs. Each actual representative is an exponent in a request at base $q+1$. The signed-power lemma gives $C, A_1, \ldots, A_8 \geq 0$, and now $H \geq 2$ is positive. The sign of B, and therefore possibly that of q, is still undetermined at this point.

6. Authenticate the first binomial digit in either radix sign. Set $n = F + t$ and $b_j = \binom{n}{j}$. For $0 \leq j \leq t$,

$$b_j \leq (F+t)^t \leq (F^2 + F)^t \leq 2^{t(F-1)}.$$

The last elementary inequality holds for $F \geq 7$. From (8),

$$\frac{2^{t(F-1)}}{Q} \leq 2^{-F-t-2}.$$

Thus the signed low part $r_* = \sum_{j<t} b_j q^j$ has $|r_*| < Q^t/32$, and $b_t < Q/32$. The binomial theorem is valid also for $q < 0$.

To see exactly what the common norm enforces, write a denominator $D = q^k$, its ratio $a = M/D$, and a normalized remainder ρ. The literal rows give $Da = M$ and $aU = MJ + \rho$. Since $a \neq 0$, $a \mid \rho$; putting $r = \rho/a$ gives $U = DJ + r$ with $r \in \mathbb{Z}$. The common norm is

$$M^2 = 1 + 4\sum_{i=1}^{11} \rho_i^2 + \alpha^2 + \beta^2 + \gamma(\gamma+1). \tag{9}$$

Its last three terms are nonnegative for every integer input. Therefore $|\rho_i| < |M|/2$ and $|r_i| < |D_i|/2$, regardless of denominator signs. For the first binomial numerator, $D = q^t$ divides $r - r_*$, while $|r - r_*| < |D|/2 + |D|/32 < |D|$. Hence $r = r_*$.

The next digit satisfies $B \equiv b_t \pmod{q}$. Put $a = |BH| \geq 2$. As $x \geq 3$, $|B| \leq a < Q/2$. Hence $|B - b_t| < Q/2 + Q/32 < Q$ and

$$B = b_t = \binom{F+t}{t} > 0. \tag{10}$$

Now $q = (BH)^x > 0$, $B \geq 2$, and the B request gives $N = B^x > 0$. This finishes the sign recovery without using the binomial conclusion to establish a sign needed in its own proof. □

7 Recovering all local rows and a refutation

The new positive H dominates the old majorant $2 + C + \sum A_j$. For an ordinary representative $a \geq 0$, every needed low coefficient satisfies $\binom{a}{j} \leq \max(1, a)^x \leq H^x$. This includes zero representatives and the constant coefficient. The inherited derived-label bound is $h_* \leq H^\ell$ with $\ell = 2^{22}$. Since $F > \ell$ and $x > (t+1)F$, its needed coefficients also lie below H^x. Since $q = B^x H^x$ and $B \geq 2$,

$$\frac{H^x}{q-1} < \frac{2}{2^x} < \frac{1}{32}.$$

Thus all ten remaining normalized binomial remainders are identified exactly by the same argument used for the first digit.

7.1 The binomial CRT implication

The elementary range fact is: if p is prime and $p^e \mid \binom{a}{k}$, with $a \geq 0$ and $k, e \geq 1$, then $a \bmod p^e < k$. If $a < k$ it is immediate. Otherwise the factorial valuation is a sum of 0-or-1 terms. If the k consecutive factors $a-k+1, \ldots, a$ contain no multiple of p^e, all terms from level e onward vanish, leaving valuation at most $e-1$, a contradiction.

Use $N = B^x$. Fix a row $i \in \{1, \ldots, t\}$ and its selector $p^e \| B$ from Lemma 5.2. The exact index range $N \mid \binom{C}{t}$ gives $C \equiv s \pmod{p^{xe}}$ with $0 \leq s < t$. The raw congruence $B \mid C + F + 1$, reduced modulo $p^e \mid F + i$, gives $s \equiv i - 1 \pmod{p^e}$. Since $p^e > t$, $s = i - 1$ exactly. The other binomial ranges similarly recover the bounded local parameters.

The unchanged local polynomial L is nonnegative and has the bound

$$L < (2 + t + x + y + F)^E, \qquad E = 2^{23}.$$

By (8) and $y \geq F$, the parenthesis is less than $2x$; also $x > 2^{32}$. The elementary dyadic estimate $E \log_2(2x) < x$ therefore gives $L < 2^x$. For example, if $2^k \leq x < 2^{k+1}$ and $k \geq 32$, then $E \log_2(2x) < 2^{23}(k+2) < 2^k \leq x$; the middle inequality holds at $k = 32$ and persists by induction. The local congruence makes p^{xe} divide L, but $p^{xe} \geq 2^x$, so $L = 0$. Every required row is valid. The derived label is a polynomial of the independent columns, so its congruence and nonnegative size bound identify the same local label.

7.2 The exceptional beta boundary and the Horn argument

The affine choice permits $y = F$ algebraically. At that boundary, the first endpoint has beta remainder 0 modulo $y + 1$, while the last has remainder 1. The decoded local updates have the form

$$S_{i+1} W_{v_i} = S_i W_{p_i} W_{q_i},$$

with every weight positive. An initial zero propagates to zero, contradicting the terminal value. Thus every actual zero has $y > F$, initial accumulator $F + 1$, and terminal accumulator 1. All accumulators are positive; multiplication and cancellation yield (6).

Lemma 4.1 gives the exact multiplicity balance. Take the finite closure of truth under the rule instances appearing in the certificate. If falsity were outside that closure, each row with an outside conclusion would have at least one outside premise. Summing conclusions minus premises over outside labels would then be nonpositive. The recovered balance makes that sum one, a contradiction. Thus there is a finite C_{222} refutation. The identified calibration gives a finite ZFC refutation. Cyclic rule multisets do not invalidate this argument; the single-conclusion property is the relevant restriction.

8 Constructive completeness

Start with a finite C_{222} refutation. Expand it to a finite proof tree and choose $t = 16 + 4h^2$ beyond its size, schema parameters and label bounds. This cofinal choice is made before the large arithmetic witnesses. Set $F = t^t$ and construct the finite pending-multiset accumulator path with weights $1 + jF$, padding with truth to exactly t rows.

Let O be the odd part of $t!$. Choose an arbitrarily large positive odd multiple w of O and put

$$y = Fw, \qquad v = (w-1)/2.$$

Since $v_2(F) = t\, v_2(t) \geq 2t > v_2(t!)$, this y is an even multiple of $t!$. Increase w so that $y > F$ and every accumulator is at most y. The moduli $1 + (i+1)y$, $0 \leq i \leq t$, are pairwise coprime: a common

prime would divide a nonzero index difference at most t, hence divide y, contrary to its divisibility into $1 + (i + 1)y$. Choose a sufficiently large positive beta code x in the required CRT class. Its local beta quotients are nonnegative and strictly below x because their denominators are at least two. The genuine endpoints give (8).

Set $B = \binom{F+t}{t}$ and $N = B^x$. To construct the second CRT stage, for *every* prime $p \mid B$ choose a compatible row using the second part of Lemma 5.2. Prescribe that row's local tuple and index modulo p^{2xe}, where $e = v_p(B)$. The pairwise coprime moduli have product N^2. Range binomials are divisible by p^{xe}: their falling factorials contain the selected factor divisible by p^{2xe}, and division by $k!$ loses at most $v_p(k!) \le k - 1 < x \le xe$. The zero-binomial case when the upper argument is smaller than k is immediate. This supplies all required global congruences, including those at unused selectors.

The binomial B is odd. Indeed $v_2(F) > \lfloor \log_2 t \rfloor$, so $v_2(F + i) = v_2(i)$ for $1 \le i \le t$; subtracting $v_2(t!)$ gives zero. Thus N^2 is odd, and adding N^2 changes a representative's parity without changing any CRT congruence. Choose C even, giving $c = C/2$. As in the parent, choose the dominant schema representative a odd in $[2N^2, 4N^2)$ and b even; choose the other representatives positive and at most $2N^2$. The inherited constructor satisfies

$$h_* = 1 + a + b \pmod 2, \qquad h_* \ge 2a^2.$$

It therefore supplies the same even dominant semantic anchor. The changed front index is smaller: $F + t \le \binom{F+t}{t} = B < N^2$. All other requested indices are bounded as in the parent construction.

Only now define $H = 2 + 4(C + \sum A_j)$ and $q = (BH)^x$. This order avoids any circular choice of a CRT modulus. Here q is even, which is allowed; the retained range argument uses $N \mid q$, since $q = NH^x$, and does not require odd q. All four binomial denominator groups divide $M = q^x$. Their genuine normalized remainders satisfy $0 \le \rho_i < M/32$, so

$$4 \sum_{i=1}^{11} \rho_i^2 < \frac{44}{1024} M^2 < M^2.$$

The remaining integer gap in (9) is nonnegative and is representable as $a^2 + b^2 + c(c + 1)$. This standard consequence of the three-square theorem follows by representing $4g + 1$: exactly one square is odd and two are even, and division by four gives the displayed form.

The retained constructive Pell frame and semantic anchor now supply the finite family of genuine power certificates. The first request still has the existing index t. The frame's duplicated base/index atom is harmless. This constructs a zero of the changed source. Exact finite transformations then give zeros of each released polynomial.

9 From the source to the literal quartics

The complete source mutation changes eleven surviving quadratic rows and deletes three rows. All eighteen power requests remain. The deleted registers are the squared radius, the private radix square root, and the beta multiplier's square wire. Historical unused slots remain in the raw array only to make the source comparison exact; they are absent from the final coordinate list.

The finite construction then uses the retained Horn factoring identity, 202 unit-monic eliminations, deletion of 117 unused slots, and 69 integral translations to obtain the 242-coordinate intermediate system with 178 quadratic rows. Six credited integer coordinate shears and exact row identities expose three affine eliminations for the sparse 239-variable system, or four for the 238-variable system. The last step is a recorded sequence of row replacements $R_i \leftarrow R_i + kR_j$, $i \ne j$, $k \in \mathbb{Z}$. These have the explicit inverse $R_i \leftarrow R_i - kR_j$ and preserve the common zero set. They change the sum-of-squares polynomial and can reduce its coefficient height.

For every unit-monic equation $\varepsilon z + f = 0$, $\varepsilon = \pm 1$, the removed coordinate is restored as the integer polynomial $z = -\varepsilon f$. The certificates are acyclic and the checker verifies that every retained row stays quadratic. The final affine transformations give a polynomial bijection of zero sets with the intermediate system. Deleting unused coordinates earlier gives existence equivalence with free fibers; it is not described as a bijection.

No expansion estimate is used for the reported metrics. Every final coefficient is collected exactly. A nonzero degree-four term establishes exact degree four, and every declared coordinate is checked to occur.

10 Verification, reproducibility and proof boundary

The archive separates three kinds of evidence.

Unbounded mathematics. The proofs in this report and the two independent derivation/review notes cover label separation, the LCM/radius inequality, the ordered signed bootstrap, binomial decoding, both completeness constructions, and the composition with the inherited logical interfaces. Those arguments, rather than a finite search, establish the quantified equivalence.

Exact finite algebra. A separately written checker imports no producer or optimizer. It replays the entire new source delta; seventeen affine-square identities and eighteen Horn rewrites with private-wire checks; every monic restoration and integral translation; and the complete final coordinate/row certificates. For each endpoint it pulls back all 178 intermediate residuals through the inverse row operations and compares every polynomial coefficient. Both ordinary and optimized Python execution pass. Targeted controls reject a weakened $C = 2c$ chart, a changed Horn coefficient despite an updated hash, and a monic restoration changed by one.

Supplementary arithmetic diagnostics. Independent finite checks cover the LCM recurrences through 500, 168 binomial-CRT examples, 756 row selectors, 1,269 full-prime support checks, 17,576 label-multiset coefficient/injectivity cases, and 4,878 unequal-factor count comparisons. They also check actual parity/beta completions and a counterexample to removing the CRT radius hypothesis. These diagnostics help catch transcription mistakes; they are not substitutes for the proofs.

Run, with Python 3.10 or newer and its standard library,

```
python3 -B verify.py
python3 -B verify.py --rebuild
python3 -B -O verify.py
```

The rebuild mode reconstructs the source and all selected coefficient files byte for byte in a temporary directory. It replays frozen row certificates; it does not rerun an optimizer or search for a refutation. To evaluate a supplied 238-integer tuple stored as a JSON list, run

```
python3 -B evaluate.py point.json
```

The evaluator is exact and makes no attempt to decide solvability.

Rebuild the standalone article with `python3 -B build_report.py`. The report source includes the complete printed polynomial definition. Its generated portions are compared with the authoritative JSON. Different TeX installations can change PDF metadata and bytes; the mathematical files and deterministic source reconstruction are the reproducibility targets.

11 Assessment and further work

The achieved result is a smaller direct quartic with a more economical arithmetic justification: one growth parameter performs two jobs, and existing powers enforce signs that previously required separate squares. The sharp point in the label proof is the positivity of both coefficient lists, which allows $F = t^t$ exactly. The sharp point in the bootstrap is using q^{y+1} before using the representative powers at base $q + 1$. Interchanging those steps would reintroduce a circular argument.

This is a concrete new reduction relative to the inspected source lineage. It is not a demonstrated breakthrough in the global minimum number of Diophantine unknowns. The most substantial remaining arithmetic cost is the shared certification of eighteen powers; a different simultaneous-power certificate could have a much larger effect than another coordinate or row-basis adjustment. The compact new lemmas and the exact source delta also provide clear targets for future formalization. No unproved shortcut is included in the released endpoints.

References

[1] V. Reshetnikov and ProveIt contributors, *ProveIt*, comparison snapshot `7a8da414376a80b90ecf799f37ceb26fb9ac674e`, 25 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/7a8da414376a80b90ecf799f37ceb26fb9ac674e/Computability/DiophantineConsistency/con-zfc-diophantine-forms.

[2] ProveIt incoming research, *Removing factorial authenticity: binomial prime-power certificates for Diophantine consistency*, `ConZFC_Factorial_Free_Binomial_CRT_2026-10-25.zip`, retained in `inherited/factorial_free/`. https://github.com/VladimirReshetnikov/ProveIt/blob/7a8da414376a80b90ecf799f37ceb26fb9ac674e/docs/incoming/ConZFC_Factorial_Free_Binomial_CRT_2026-10-25.zip.

[3] ProveIt incoming research, *Shared factorial errors: explicit Diophantine equations for Con(ZFC)*, `ConZFC_Shared_Factorial_Error_2026-10-25.zip`. https://github.com/VladimirReshetnikov/ProveIt/blob/7a8da414376a80b90ecf799f37ceb26fb9ac674e/docs/incoming/ConZFC_Shared_Factorial_Error_2026-10-25.zip.

[4] B. Farhi, *An identity involving the least common multiple of binomial coefficients and its application*, 2009, arXiv:0906.2295v2. https://arxiv.org/abs/0906.2295.

[5] J. P. Jones, *Universal Diophantine equation*, Journal of Symbolic Logic **47**(3) (1982), 549–571. https://doi.org/10.2307/2273588.

A Complete principal polynomial

All coordinates $x_0, \ldots, x_{237}$ are unrestricted integers. The single equation is exactly

$$\sum_{i=0}^{173} R_i(x_0, \ldots, x_{237})^2 = 0.$$

The following list is complete. Symbols in this list are final coordinates; their interpretations in the arithmetic construction are obtained through the supplied restoration certificates. There are no coefficient ports or additional implicit unknowns.

$$R_0 = x_0 - x_0\,x_2 - x_2 + x_7$$

$$
\begin{aligned}
R_{1} &= -x_{2}\,x_{5} - x_{3} + x_{6}\,x_{7} \\
R_{2} &= -1 + x_{1} - x_{6}\,x_{7} \\
R_{3} &= 1 + x_{3} - x_{4}\,x_{18} + 2x_{16} \\
R_{4} &= -1 - x_{8}^{2} - 2x_{8}\,x_{9} - x_{9}^{2} + x_{19} \\
R_{5} &= -10x_{8} + x_{8}^{2} - 6x_{8}\,x_{9} - 8x_{8}\,x_{19} - 20x_{9} \\
&\quad - 11x_{9}^{2} - 16x_{9}\,x_{19} - 15x_{19} - 4x_{19}^{2} + x_{20} \\
R_{6} &= -2 + 2x_{8} - x_{8}^{2} - 4x_{8}\,x_{20} - 4x_{20} \\
&\quad - 4x_{20}^{2} + x_{21} \\
R_{7} &= -24x_{8}\,x_{21} + 24x_{8}\,x_{35} + 2x_{21} - 4x_{21}^{2} + x_{22} \\
&\quad - 2x_{35} + 4x_{35}^{2} - x_{36} \\
R_{8} &= -2 - x_{9} - x_{9}^{2} - 2x_{9}\,x_{10} - x_{10} \\
&\quad - x_{10}^{2} + x_{23} \\
R_{9} &= -8 + 7x_{8} - x_{8}^{2} + 4x_{8}\,x_{10} - 4x_{8}\,x_{23} \\
&\quad - 4x_{10} + 4x_{23} + x_{24} - x_{46} \\
R_{10} &= -1 + x_{8} - x_{8}^{2} - 2x_{8}\,x_{10} + x_{10} \\
&\quad - x_{10}^{2} + x_{25} \\
R_{11} &= 17 + 4x_{8} - 8x_{8}\,x_{9} - 8x_{8}\,x_{10} - 16x_{8}\,x_{19} \\
&\quad - 8x_{8}\,x_{25} + 19x_{9} + 9x_{9}^{2} - 36x_{9}\,x_{10} - 12x_{9}\,x_{25} \\
&\quad + 8x_{10} + 8x_{10}^{2} - 24x_{10}\,x_{19} - 8x_{19}\,x_{25} - x_{20} \\
&\quad - 4x_{25} + x_{26} - x_{31} - 8x_{33} - 8x_{33}^{2} \\
&\quad + 2x_{34} \\
R_{12} &= -4x_{24}^{2} - 8x_{24}\,x_{26} - 4x_{26}^{2} + x_{27} \\
R_{13} &= -2 - 8x_{8} + 8x_{8}\,x_{23} + 8x_{8}\,x_{25} - 8x_{9} \\
&\quad + 8x_{9}\,x_{23} + 8x_{9}\,x_{25} - 4x_{19} + 8x_{23} - 4x_{23}^{2} \\
&\quad - 8x_{23}\,x_{25} + 8x_{25} - 4x_{25}^{2} + x_{28} \\
R_{14} &= 6 - 6x_{8} + x_{8}^{2} + 26x_{8}\,x_{9} + 8x_{8}\,x_{19} \\
&\quad + 16x_{8}\,x_{25} + 8x_{8}\,x_{28} + 4x_{9} - 11x_{9}^{2} - 16x_{9}\,x_{19} \\
&\quad - 32x_{9}\,x_{25} - 16x_{9}\,x_{28} - 3x_{19} - 4x_{19}^{2} - 16x_{19}\,x_{25} \\
&\quad - 8x_{19}\,x_{28} + 8x_{25} - 16x_{25}^{2} - 16x_{25}\,x_{28} + 4x_{28} \\
&\quad - 4x_{28}^{2} + x_{29} \\
R_{15} &= 6 + x_{0} - 5x_{8} - x_{8}^{2} - 8x_{8}\,x_{9} \\
&\quad - 12x_{8}\,x_{10} - 4x_{8}\,x_{23} - 8x_{9} - 16x_{9}\,x_{10} - 16x_{9}\,x_{23} \\
&\quad - 20x_{10} - 20x_{10}^{2} - 24x_{10}\,x_{23} - 28x_{23} - 4x_{23}^{2} \\
&\quad + x_{30} - 4x_{213}^{2} \\
R_{16} &= 3 - 15x_{8} + 13x_{8}^{2} - 8x_{8}\,x_{9} - 14x_{8}\,x_{10} \\
&\quad + 8x_{8}\,x_{25} + 24x_{8}\,x_{39} + 3x_{9} + 5x_{9}^{2} - 4x_{9}\,x_{25} \\
&\quad - 15x_{10} - 21x_{10}^{2} - 8x_{10}\,x_{25} + 16x_{10}\,x_{39} - 6x_{23} \\
&\quad - 3x_{25} + 8x_{25}\,x_{39} + x_{31} - 4x_{39} + 4x_{39}^{2} \\
&\quad - x_{40} \\
R_{17} &= 2 - 4x_{30}^{2} - 8x_{30}\,x_{31} - 4x_{31}^{2} + x_{32}
\end{aligned}
$$

$$
\begin{aligned}
R_{18} = {} & 1 - 7x_8 + 12x_8^2 + 12x_8\,x_9 + 4x_8\,x_{10} \\
& - 8x_8\,x_{25} - 8x_8\,x_{63} - 26x_9 - 8x_9^2 + 4x_9\,x_{10} \\
& + 8x_9\,x_{19} + 16x_9\,x_{23} - 8x_9\,x_{25} - 16x_9\,x_{63} - 6x_{10} \\
& - 4x_{10}^2 + 4x_{19} - 8x_{19}\,x_{23} + 8x_{19}\,x_{63} + 6x_{23} \\
& + 8x_{23}\,x_{25} - 9x_{25} + 4x_{25}^2 - x_{28} + x_{33} \\
& - 4x_{63} + 4x_{63}^2 - x_{64} + 4x_{76}
\end{aligned}
$$

$$
\begin{aligned}
R_{19} = {} & - 3 - 6x_8 - 4x_{10} - 2x_{25} - 4x_{33} \\
& - 4x_{33}^2 + x_{34}
\end{aligned}
$$

$$
\begin{aligned}
R_{20} = {} & 7 + 16x_8 + 7x_8^2 - 4x_8\,x_{10} + 28x_8\,x_{20} \\
& + 8x_8\,x_{23} + 8x_8\,x_{25} + 8x_8\,x_{63} + 28x_9 - 32x_9^2 \\
& - 8x_9\,x_{10} - 32x_9\,x_{19} + 64x_9\,x_{20} + 8x_9\,x_{23} + 8x_9\,x_{25} \\
& + 16x_9\,x_{63} - 7x_{10} + x_{10}^2 + 4x_{10}\,x_{19} + 18x_{19} \\
& - 24x_{19}^2 + 32x_{19}\,x_{20} - 8x_{19}\,x_{63} + 14x_{20} - 4x_{20}^2 \\
& + 8x_{23} - 4x_{23}^2 - 8x_{23}\,x_{25} + 4x_{25} - 4x_{25}^2 \\
& + x_{28} - 8x_{33} - 8x_{33}^2 + 2x_{34} + x_{35} \\
& + x_{38} + 3x_{63} - 4x_{63}^2 + x_{64}
\end{aligned}
$$

$$
\begin{aligned}
R_{21} = {} & 1 + 6x_8 - 36x_8^2 - 24x_8\,x_{35} + 2x_{35} \\
& - 4x_{35}^2 + x_{36}
\end{aligned}
$$

$$
\begin{aligned}
R_{22} = {} & - x_9^2 - 4x_9\,x_{36} - 4x_{36}^2 + x_{37}
\end{aligned}
$$

$$
\begin{aligned}
R_{23} = {} & - 6 - 2x_8 - 4x_9 - 14x_{19} - 16x_{19}^2 \\
& + x_{38}
\end{aligned}
$$

$$
\begin{aligned}
R_{24} = {} & 8 + x_0 + 21x_9 - 21x_9^2 - 18x_9\,x_{10} \\
& - 24x_9\,x_{23} - 24x_9\,x_{38} + 3x_{10} - 9x_{10}^2 - 16x_{10}\,x_{38} \\
& + 3x_{23} - 12x_{23}^2 - 8x_{23}\,x_{38} + 4x_{38} - 4x_{38}^2 \\
& + x_{39} + 2x_{46} - 4x_{213}^2
\end{aligned}
$$

$$
\begin{aligned}
R_{25} = {} & 16 - 10x_8 - 20x_8^2 - 16x_8\,x_{10} - 24x_8\,x_{25} \\
& - 24x_8\,x_{39} - 12x_{10} - 16x_{10}\,x_{25} - 16x_{10}\,x_{39} - 14x_{25} \\
& - 4x_{25}^2 - 8x_{25}\,x_{39} + 4x_{39} - 4x_{39}^2 + x_{40}
\end{aligned}
$$

$$
\begin{aligned}
R_{26} = {} & - x_{10}^2 - 2x_{10}\,x_{11} - x_{11}^2 + x_{41}
\end{aligned}
$$

$$
\begin{aligned}
R_{27} = {} & 2 + x_8^2 - 4x_8\,x_{41} + 2x_{10} - 2x_{10}^2 \\
& - 8x_{10}\,x_{41} - 2x_{25} + 4x_{41} - 4x_{41}^2 + x_{42}
\end{aligned}
$$

$$
\begin{aligned}
R_{28} = {} & 1 - 2x_0 - 16x_{10} - 16x_{10}\,x_{23} + 16x_{10}\,x_{25} \\
& - 8x_{23} + 12x_{23}^2 - 8x_{23}\,x_{25} + 24x_{25} - 4x_{25}^2 \\
& + x_{43} - 4x_{46} + 8x_{213}^2
\end{aligned}
$$

$$
\begin{aligned}
R_{29} = {} & - 4x_{10} - 16x_{25}^2 - 32x_{25}\,x_{41} - 16x_{25}\,x_{42} - 16x_{25}\,x_{43} \\
& - 4x_{41} - 16x_{41}^2 - 16x_{41}\,x_{42} - 16x_{41}\,x_{43} - 2x_{42} \\
& - 4x_{42}^2 - 8x_{42}\,x_{43} - 4x_{43}^2 + x_{44}
\end{aligned}
$$

$$
\begin{aligned}
R_{30} = {} & - 16x_8^2 + 8x_8\,x_{10} + 32x_8\,x_{25} + 16x_8\,x_{28} + x_{10} \\
& - x_{10}^2 - 8x_{10}\,x_{25} - 4x_{10}\,x_{28} - 16x_{25}^2 - 16x_{25}\,x_{28} \\
& - 4x_{28}^2 + x_{45}
\end{aligned}
$$

$$
\begin{aligned}
R_{31} = & -1 - 8x_{10} - 4x_{10}^2 + 8x_{10}\,x_{23} + 8x_{23} \\
& - 4x_{23}^2 + x_{46}
\end{aligned}
$$

$$
\begin{aligned}
R_{32} = & -1 - 9x_{10}^2 + 24x_{10}\,x_{23} + 12x_{10}\,x_{46} - 16x_{23}^2 \\
& - 16x_{23}\,x_{46} - 4x_{46}^2 + x_{47}
\end{aligned}
$$

$$
\begin{aligned}
R_{33} = & 1 + x_9 - x_9^2 - 4x_9\,x_{10} - 4x_9\,x_{47} \\
& - 4x_{10}^2 - 8x_{10}\,x_{47} - 4x_{47}^2 + x_{48}
\end{aligned}
$$

$$
R_{34} = -1 - 4x_{48} - 4x_{48}^2 + x_{49}
$$

$$
\begin{aligned}
R_{35} = & -1 - x_{10}^2 - 8x_{10}\,x_{48} - 4x_{10}\,x_{49} - 16x_{48}^2 \\
& - 16x_{48}\,x_{49} - 4x_{49}^2 + x_{50}
\end{aligned}
$$

$$
\begin{aligned}
R_{36} = & 2 + x_9 - x_9^2 - 4x_9\,x_{10} - 4x_9\,x_{50} \\
& - 4x_{10}^2 - 8x_{10}\,x_{50} - 4x_{50}^2 + x_{51}
\end{aligned}
$$

$$
R_{37} = -4x_{45}^2 - 8x_{45}\,x_{51} - 4x_{51}^2 + x_{52}
$$

$$
\begin{aligned}
R_{38} = & -9 + 7x_8 - x_8^2 - 8x_8\,x_{45} - 4x_8\,x_{52} \\
& + 24x_{45} - 16x_{45}^2 - 16x_{45}\,x_{52} + 12x_{52} - 4x_{52}^2 \\
& + x_{53}
\end{aligned}
$$

$$
R_{39} = -1 - 4x_{53} - 4x_{53}^2 + x_{54}
$$

$$
\begin{aligned}
R_{40} = & -x_{10}^2 - 8x_{10}\,x_{53} - 4x_{10}\,x_{54} - 16x_{53}^2 - 16x_{53}\,x_{54} \\
& - 4x_{54}^2 + x_{55}
\end{aligned}
$$

$$
\begin{aligned}
R_{41} = & -x_8 - x_8^2 - 4x_8\,x_{10} - 4x_8\,x_{55} - 4x_{10} \\
& - 4x_{10}^2 - 8x_{10}\,x_{55} - 4x_{55} - 4x_{55}^2 + x_{56}
\end{aligned}
$$

$$
R_{42} = -10 - 12x_{56} - 4x_{56}^2 + x_{57}
$$

$$
\begin{aligned}
R_{43} = & 2 + 4x_{10} + 16x_{10}\,x_{56} + 8x_{10}\,x_{57} - 4x_{23} \\
& - 16x_{23}\,x_{56} - 8x_{23}\,x_{57} - x_{46} + 8x_{56} - 16x_{56}^2 \\
& - 16x_{56}\,x_{57} + 4x_{57} - 4x_{57}^2 + x_{58}
\end{aligned}
$$

$$
\begin{aligned}
R_{44} = & 1 + x_9 - x_9^2 + 8x_9\,x_{10} - 8x_9\,x_{23} \\
& - 4x_9\,x_{58} - 7x_{10}^2 + 8x_{10}\,x_{23} - 12x_{10}\,x_{46} + 16x_{10}\,x_{58} \\
& + 16x_{23}\,x_{46} - 16x_{23}\,x_{58} + 4x_{46}^2 - x_{47} - 4x_{58}^2 \\
& + x_{59}
\end{aligned}
$$

$$
R_{45} = -10 - 12x_{51} - 4x_{51}^2 + x_{60}
$$

$$
\begin{aligned}
R_{46} = & 1 - 16x_{51}^2 - 16x_{51}\,x_{59} - 16x_{51}\,x_{60} - 4x_{59}^2 \\
& - 8x_{59}\,x_{60} - 4x_{60}^2 + x_{61}
\end{aligned}
$$

$$
\begin{aligned}
R_{47} = & x_{10} - x_{10}^2 - 8x_{10}\,x_{59} - 4x_{10}\,x_{61} - 16x_{59}^2 \\
& - 16x_{59}\,x_{61} - 4x_{61}^2 + x_{62}
\end{aligned}
$$

$$
\begin{aligned}
R_{48} = & 4x_8 - 4x_8^2 - 12x_8\,x_9 + 8x_8\,x_{19} + 4x_9 \\
& - 16x_9^2 + 4x_9\,x_{10} + 16x_9\,x_{19} + 3x_{10} + 3x_{10}^2 \\
& - 4x_{10}\,x_{19} - 4x_{19} - 4x_{19}^2 + x_{63} - 4x_{76}
\end{aligned}
$$

$$
\begin{aligned}
R_{49} = & 1 - 8x_8 - 4x_8\,x_{10} + 8x_8\,x_{63} - 16x_9 \\
& - 8x_9\,x_{10} + 16x_9\,x_{63} + x_{10} + x_{10}^2 + 4x_{10}\,x_{19} \\
& + 8x_{19} - 8x_{19}\,x_{63} + 3x_{63} - 4x_{63}^2 + x_{64}
\end{aligned}
$$

$$
\begin{aligned}
R_{50} = {} & -3 - 4x_8 - 12x_8\,x_{10} + 16x_8\,x_{62} - 8x_8\,x_{63} \\
& + 16x_8\,x_{64} - 8x_9 - 24x_9\,x_{10} + 32x_9\,x_{62} - 16x_9\,x_{63} \\
& + 32x_9\,x_{64} + 3x_{10} + 3x_{10}^2 + 12x_{10}\,x_{19} + 4x_{19} \\
& - 16x_{19}\,x_{62} + 8x_{19}\,x_{63} - 16x_{19}\,x_{64} - 4x_{62}^2 - 8x_{62}\,x_{64} \\
& - 7x_{63} + 4x_{63}^2 - 3x_{64} - 4x_{64}^2 + x_{65}
\end{aligned}
$$

$$R_{51} = -2 - 4x_{65} - 4x_{65}^2 + x_{66}$$

$$R_{52} = -x_3\,x_{13} + x_{67}$$

$$R_{53} = x_{12} - x_{13} - 2x_{22}\,x_{67} - x_{68}$$

$$R_{54} = x_{12} - x_{13} - 4x_{24}\,x_{67} - 2x_{27}\,x_{67} - x_{69}$$

$$R_{55} = -2x_{29}\,x_{67} + 2x_{44}\,x_{67} - 2x_{67} - x_{70} + x_{75}$$

$$R_{56} = x_{12} - x_{13} - 4x_{30}\,x_{67} - 2x_{32}\,x_{67} - x_{71}$$

$$R_{57} = x_{12} - x_{13} - 2x_{34}\,x_{67} + 6x_{67} - x_{72}$$

$$R_{58} = x_{12} - x_{13} - 4x_{36}\,x_{67} - 2x_{37}\,x_{67} - x_{73}$$

$$R_{59} = x_{12} - x_{13} - 2x_{40}\,x_{67} - x_{74}$$

$$R_{60} = x_{12} - x_{13} - 2x_{44}\,x_{67} - 4x_{67} - x_{75}$$

$$
\begin{aligned}
R_{61} = {} & -x_8\,x_9 + x_8\,x_{10} + x_9 + x_9\,x_{10} - x_{10} \\
& - x_{10}^2 + x_{76}
\end{aligned}
$$

$$R_{62} = 1 + 2x_{11} + x_{11}^2 - x_{76}^2 + x_{77}$$

$$R_{63} = -x_{77}^2 + x_{78}$$

$$R_{64} = -4x_{65}\,x_{67} - 2x_{66}\,x_{67} + 2x_{67} + x_{79}$$

$$
\begin{aligned}
R_{65} = {} & -x_{12}^2 + 2x_{12}\,x_{13} + 2x_{12}\,x_{79} - x_{13}^2 - 2x_{13}\,x_{79} \\
& - x_{79}^2 + x_{80}
\end{aligned}
$$

$$R_{66} = -1 - x_3\,x_9 + x_{81}$$

$$R_{67} = -x_{13}\,x_{81} + x_{83}\,x_{84} + x_{85}$$

$$R_{68} = -1 + x_3 - x_3\,x_8 + x_{82}$$

$$
\begin{aligned}
R_{69} = {} & -1 + 2x_3 + 2x_3\,x_8 + 4x_3\,x_9 - 2x_3\,x_{19} \\
& + x_{83}
\end{aligned}
$$

$$R_{70} = -x_{12}\,x_{82} + x_{84}$$

$$
\begin{aligned}
R_{71} = {} & -6x_3\,x_{13} + x_{12}\,x_{81} + 12x_{13} - 8x_{13}\,x_{81} - 4x_{13}\,x_{82} \\
& - x_{13}\,x_{83} + x_{86}
\end{aligned}
$$

$$R_{72} = -x_{12}\,x_{68} + x_{13}\,x_{68} + x_{87}$$

$$R_{73} = -x_{69}\,x_{87} + x_{88}$$

$$R_{74} = -x_{70}\,x_{88} + x_{89}$$

$$R_{75} = -x_{71}\,x_{89} + x_{90}$$

$$R_{76} = -x_{72}\,x_{90} + x_{91}$$

$$R_{77} = -x_{73}\,x_{91} + x_{92}$$

$$R_{78} = -x_{74}\,x_{92} + x_{93}$$

$$R_{79} = -x_{75}\,x_{93} + x_{94}$$

$$R_{80} = -x_{85}\,x_{94} + x_{95}$$

$$R_{81} = -x_{86}\,x_{95} + x_{96}$$

$$R_{82} = -x_{96}^2 + x_{97}$$

$$R_{83} = -x_2 - 2x_2\,x_{16} + 2x_{16} + x_{98}$$

$$R_{84} = -x_{14}\,x_{98} + x_{99}$$

$$\begin{aligned} R_{85} = {} & -2x_1^2 + 2x_1\,x_{12} + 2x_1\,x_{13} + 2x_1\,x_{99} + 2x_1\,x_{100} \\ & - x_{12}^2 - 2x_{12}\,x_{99} - x_{13}^2 - 2x_{13}\,x_{100} + x_{17}\,x_{101} \\ & - x_{78}\,x_{97} - x_{80}\,x_{97} - x_{99}^2 - x_{100}^2 \end{aligned}$$

$$R_{86} = -x_2\,x_{15} + x_{15} - x_{15}\,x_{98} + x_{100}$$

$$R_{87} = -x_{102}\,x_{105} + x_{105} + x_{107}$$

$$R_{88} = -x_4\,x_{129} + x_{103}\,x_{228} - x_{106} - x_{107}\,x_{129}$$

$$R_{89} = x_{108}\,x_{229} - x_{110}\,x_{132} - x_{111}$$

$$R_{90} = x_{112}\,x_{229} - x_{113}\,x_{132} - x_{114}$$

$$R_{91} = x_{115}\,x_{229} - x_{116}\,x_{132} - x_{117}$$

$$R_{92} = x_{118}\,x_{229} - x_{119}\,x_{132} - x_{120}$$

$$R_{93} = x_{121}\,x_{230} - x_{123}\,x_{132} - x_{124}$$

$$R_{94} = x_{125}\,x_{230} - x_{126}\,x_{132} - x_{127}$$

$$R_{95} = -x_{17}\,x_{129} + x_{132}$$

$$R_{96} = x_{128} - x_{130}\,x_{132} - x_{131}$$

$$R_{97} = -x_{132}\,x_{134} + x_{133} - x_{135}$$

$$R_{98} = -x_{132}\,x_{137} + x_{136}\,x_{228} - x_{138}$$

$$R_{99} = -x_{132}\,x_{140} + x_{139}\,x_{229} - x_{141}$$

$$R_{100} = x_{142} - 2x_{142}\,x_{144} + x_{145}$$

$$R_{101} = -x_{142}\,x_{145} - x_{145} + x_{146}$$

$$R_{102} = x_{145}^2 - x_{146}^2 + x_{166}$$

$$R_{103} = -x_{143}\,x_{148} - x_{148} + x_{149}$$

$$R_{104} = 2x_{147} + x_{147}^2 + x_{148}^2 - x_{149}^2$$

$$R_{105} = 1 - 2x_0 + x_0^2 - 2x_0\,x_{143} + x_{150}$$

$$\begin{aligned} R_{106} = {} & -1 - x_0\,x_{148} + x_3 + x_{143}\,x_{148} - x_{147} \\ & + x_{148} + x_{150}\,x_{151} \end{aligned}$$

$$R_{107} = 1 - 2x_4 + x_4^2 - 2x_4\,x_{143} + x_{152}$$

$$R_{108} = -x_4\,x_{197} + x_{17} + x_{152}\,x_{153} - x_{196} + x_{198}$$

$$R_{109} = -1 + 2x_{143}\,x_{212} - x_{154} + 2x_{212} - x_{212}^2$$

$$\begin{aligned} R_{110} = {} & -1 + x_{102} + x_{154}\,x_{155} - x_{196} - x_{197}\,x_{212} \\ & + x_{198} \end{aligned}$$

$$R_{111} = -x_{143}\,x_{157} - x_{157} + x_{158}$$

$$R_{112} = 2x_{156} + x_{156}^2 + x_{157}^2 - x_{158}^2$$

$$R_{113} = 1 - 2x_{102} + x_{102}^2 - 2x_{102}\,x_{143} + x_{159}$$

$$R_{114} = -1 - x_{102}\,x_{157} + x_{103} - x_{156} + x_{158} + x_{159}\,x_{160}$$

$$R_{115} = -1 - x_{102}\,x_{148} + 2x_{102}\,x_{161} + x_{104} - 2x_{143}\,x_{161} - x_{147} + x_{148} + x_{149} + x_{159}\,x_{161} - 3x_{161}$$

$$R_{116} = 1 + x_{143} - x_{143}\,x_{163} - x_{163} + x_{164}$$

$$R_{117} = 1 + 2x_{162} + x_{162}^2 - 2x_{163} + x_{163}^2 - x_{164}^2$$

$$R_{118} = -1 + x_{102} - x_{102}\,x_{163} + x_{108} + x_{159}\,x_{165} - x_{162} + x_{164}$$

$$R_{119} = -2x_{143} - x_{143}^2 + x_{166}$$

$$R_{120} = 1 - x_{102} - x_{102}\,x_{147} - x_{102}\,x_{149} + 2x_{102}\,x_{167} + x_{109} + x_{143}\,x_{149} - 2x_{143}\,x_{167} + x_{147} - x_{148}\,x_{166} + 2x_{149} + x_{159}\,x_{167} - 3x_{167}$$

$$R_{121} = -x_{143}\,x_{169} - x_{169} + x_{170}$$

$$R_{122} = 2x_{168} + x_{168}^2 + x_{169}^2 - x_{170}^2$$

$$R_{123} = -1 - x_{102}\,x_{169} + x_{112} + x_{159}\,x_{171} - x_{168} + x_{170}$$

$$R_{124} = -x_{143}\,x_{173} - x_{173} + x_{174}$$

$$R_{125} = 2x_{172} + x_{172}^2 + x_{173}^2 - x_{174}^2$$

$$R_{126} = -1 - x_{102}\,x_{173} + x_{115} + x_{159}\,x_{175} - x_{172} + x_{174}$$

$$R_{127} = -x_{143}\,x_{177} - x_{177} + x_{178}$$

$$R_{128} = 2x_{176} + x_{176}^2 + x_{177}^2 - x_{178}^2$$

$$R_{129} = -1 - x_{102}\,x_{177} + x_{118} + x_{159}\,x_{179} - x_{176} + x_{178}$$

$$R_{130} = -x_{143}\,x_{181} - x_{181} + x_{182}$$

$$R_{131} = 2x_{180} + x_{180}^2 + x_{181}^2 - x_{182}^2$$

$$R_{132} = -1 - x_{102}\,x_{181} + x_{121} + x_{159}\,x_{183} - x_{180} + x_{182}$$

$$R_{133} = -x_{143}\,x_{185} - x_{185} + x_{186}$$

$$R_{134} = 2x_{184} + x_{184}^2 + x_{185}^2 - x_{186}^2$$

$$R_{135} = -1 - x_{102}\,x_{185} + 2x_{102}\,x_{187} + x_{122} - 2x_{143}\,x_{187} + x_{159}\,x_{187} - x_{184} + x_{185} + x_{186} - 3x_{187}$$

$$R_{136} = -x_{143}\,x_{189} - x_{189} + x_{190}$$

$$R_{137} = 2x_{188} + x_{188}^2 + x_{189}^2 - x_{190}^2$$

$$R_{138} = -1 - x_{102}\,x_{189} + x_{125} + x_{159}\,x_{191} - x_{188} + x_{190}$$

$$R_{139} = -x_{143}\,x_{193} - x_{193} + x_{194}$$

$$R_{140} = 2x_{192} + x_{192}^2 + x_{193}^2 - x_{194}^2$$

$$
\begin{aligned}
R_{141} = & -1 - x_{102}\,x_{193} + x_{128} + x_{159}\,x_{195} - x_{192} \\
& + x_{194}
\end{aligned}
$$

$$
R_{142} = 1 - x_{143}\,x_{197} - x_{197} + x_{198}
$$

$$
\begin{aligned}
R_{143} = & -1 + 2x_{196} + x_{196}^2 + x_{197}^2 - 2x_{198} \\
& - x_{198}^2
\end{aligned}
$$

$$
\begin{aligned}
R_{144} = & -x_{102}\,x_{197} + 2x_{102}\,x_{199} + x_{129} - 2x_{143}\,x_{199} + x_{159}\,x_{199} \\
& - x_{196} + x_{197} + x_{198} - 3x_{199}
\end{aligned}
$$

$$
R_{145} = -x_{143}\,x_{201} - x_{201} + x_{202}
$$

$$
R_{146} = 2x_{200} + x_{200}^2 + x_{201}^2 - x_{202}^2
$$

$$
\begin{aligned}
R_{147} = & -1 - x_{102}\,x_{201} + x_{133} + x_{159}\,x_{203} - x_{200} \\
& + x_{202}
\end{aligned}
$$

$$
R_{148} = -x_{143}\,x_{205} - x_{205} + x_{206}
$$

$$
R_{149} = 2x_{204} + x_{204}^2 + x_{205}^2 - x_{206}^2
$$

$$
\begin{aligned}
R_{150} = & -1 - x_{102}\,x_{205} + x_{136} + x_{159}\,x_{207} - x_{204} \\
& + x_{206}
\end{aligned}
$$

$$
R_{151} = -x_{143}\,x_{209} - x_{209} + x_{210}
$$

$$
R_{152} = 2x_{208} + x_{208}^2 + x_{209}^2 - x_{210}^2
$$

$$
\begin{aligned}
R_{153} = & -1 - x_{102}\,x_{209} + x_{139} + x_{159}\,x_{211} - x_{208} \\
& + x_{210}
\end{aligned}
$$

$$
\begin{aligned}
R_{154} = & -2 - 2x_0 - 4x_0^2 - 2x_0\,x_3 - x_1^2 \\
& - x_2^2 - 2x_3^2 - x_4^2 + 2x_8 + 2x_8\,x_9 \\
& - x_{10}^2 - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 \\
& - x_{15}^2 - 4x_{16}^2 - x_{17}^2 - x_{19} + 2x_{102} \\
& - 2x_{102}^2 - x_{103}^2 - x_{104}^2 - x_{108}^2 - x_{109}^2 \\
& - x_{112}^2 - x_{115}^2 - x_{118}^2 - x_{121}^2 - x_{122}^2 \\
& - x_{125}^2 - x_{128}^2 - x_{129}^2 - x_{133}^2 - x_{136}^2 \\
& - x_{139}^2 + x_{142} + x_{148}^2 + x_{157}^2 - 2x_{163} \\
& + x_{163}^2 + x_{169}^2 + x_{173}^2 + x_{177}^2 + x_{181}^2 \\
& + x_{185}^2 + x_{189}^2 + x_{193}^2 + x_{197}^2 + x_{201}^2 \\
& + x_{205}^2 - x_{209}^2 - x_{212}^2 + x_{231}^2 + x_{232}^2 \\
& + x_{233} + x_{233}^2
\end{aligned}
$$

$$
\begin{aligned}
R_{155} = & 2x_4 - 4x_4\,x_8 - 4x_4\,x_9 - 4x_4\,x_{10} - 4x_4\,x_{11} \\
& - 4x_4\,x_{12} - 4x_4\,x_{13} - 4x_4\,x_{14} - 4x_4\,x_{15} - 8x_4\,x_{16} \\
& + x_{212}
\end{aligned}
$$

$$
R_{156} = -16 + x_0 - 4x_{213}^2
$$

$$
R_{157} = -x_{143}\,x_{215} - x_{215} + x_{224}
$$

$$
R_{158} = -1 + x_{214}^2 + x_{215}^2 - x_{224}^2
$$

$$
R_{159} = -x_{217} - x_{217}\,x_{218} + x_{225}
$$

$$
R_{160} = -1 + x_{216}^2 + x_{217}^2 - x_{225}^2
$$

$$R_{161} = -x_{209}^2 + x_{226}$$

$$R_{162} = x_{215} - 2x_{220}\,x_{226} - x_{226}$$

$$R_{163} = -4x_{215} - 8x_{215}\,x_{219} + x_{218}$$

$$R_{164} = -x_{143} - x_{214}\,x_{221} + x_{218}$$

$$R_{165} = -1 - x_{208} - x_{214}\,x_{222} + x_{216}$$

$$\begin{aligned} R_{166} = &-x_8 - x_9 - 4x_{19} - 2x_{22} - 4x_{24} \\ &- 2x_{27} - 2x_{29} - 4x_{30} - 2x_{32} - 2x_{34} \\ &- 4x_{36} - 2x_{37} - 2x_{40} - 2x_{44} - 4x_{65} \\ &- 2x_{66} - 4x_{209}\,x_{223} + x_{217} \end{aligned}$$

$$\begin{aligned} R_{167} = &-2 + 2x_0^2 + 2x_0\,x_3 - 2x_0\,x_{148} - 2x_0\,x_{157} \\ &+ x_1^2 - 2x_1\,x_{197} + x_2^2 - 2x_2\,x_{185} + x_3^2 \\ &- 2x_3\,x_{157} + x_8^2 - 2x_8\,x_{163} - 2x_8\,x_{209} + x_9^2 \\ &- 2x_9\,x_{169} - 2x_9\,x_{209} + x_{10}^2 - 2x_{10}\,x_{173} + x_{11}^2 \\ &- 2x_{11}\,x_{177} + x_{12}^2 - 2x_{12}\,x_{181} + x_{13}^2 - 2x_{13}\,x_{189} \\ &+ x_{14}^2 - 2x_{14}\,x_{193} + x_{15}^2 - 2x_{15}\,x_{201} + 4x_{16}^2 \\ &- 4x_{16}\,x_{205} - 8x_{19}\,x_{209} - 4x_{22}\,x_{209} - 8x_{24}\,x_{209} - 4x_{27}\,x_{209} \\ &- 4x_{29}\,x_{209} - 8x_{30}\,x_{209} - 4x_{32}\,x_{209} - 4x_{34}\,x_{209} - 8x_{36}\,x_{209} \\ &- 4x_{37}\,x_{209} - 4x_{40}\,x_{209} - 4x_{44}\,x_{209} - 8x_{65}\,x_{209} - 4x_{66}\,x_{209} \\ &- x_{143}\,x_{227} + x_{147}^2 + x_{156}^2 + x_{162}^2 + 2x_{163} \\ &+ x_{168}^2 + x_{172}^2 + x_{176}^2 + x_{180}^2 + x_{184}^2 \\ &+ x_{188}^2 + x_{192}^2 + x_{196}^2 + x_{200}^2 + x_{204}^2 \\ &+ x_{208}^2 + 2x_{209}^2 - x_{231}^2 - x_{232}^2 - x_{233} \\ &- x_{233}^2 \end{aligned}$$

$$R_{168} = -x_{104}\,x_{228} + x_{129}$$

$$R_{169} = -x_{109}\,x_{229} + x_{129}$$

$$R_{170} = -x_{122}\,x_{230} + x_{129}$$

$$R_{171} = -1 + x_2 - x_3 - 2x_3\,x_{237}$$

$$
\begin{aligned}
R_{172} = \; & -1 - 8x_8\,x_{19} - 4x_8\,x_{22} - 8x_8\,x_{24} - 4x_8\,x_{27} \\
& - 4x_8\,x_{29} - 8x_8\,x_{30} - 4x_8\,x_{32} - 4x_8\,x_{34} - 8x_8\,x_{36} \\
& - 4x_8\,x_{37} - 4x_8\,x_{40} - 4x_8\,x_{44} - 8x_8\,x_{65} - 4x_8\,x_{66} \\
& - 8x_9\,x_{19} - 4x_9\,x_{22} - 8x_9\,x_{24} - 4x_9\,x_{27} - 4x_9\,x_{29} \\
& - 8x_9\,x_{30} - 4x_9\,x_{32} - 4x_9\,x_{34} - 8x_9\,x_{36} - 4x_9\,x_{37} \\
& - 4x_9\,x_{40} - 4x_9\,x_{44} - 8x_9\,x_{65} - 4x_9\,x_{66} - x_{19} \\
& - 16x_{19}^2 - 16x_{19}\,x_{22} - 32x_{19}\,x_{24} - 16x_{19}\,x_{27} - 16x_{19}\,x_{29} \\
& - 32x_{19}\,x_{30} - 16x_{19}\,x_{32} - 16x_{19}\,x_{34} - 32x_{19}\,x_{36} - 16x_{19}\,x_{37} \\
& - 16x_{19}\,x_{40} - 16x_{19}\,x_{44} - 32x_{19}\,x_{65} - 16x_{19}\,x_{66} - 4x_{22}^2 \\
& - 16x_{22}\,x_{24} - 8x_{22}\,x_{27} - 8x_{22}\,x_{29} - 16x_{22}\,x_{30} - 8x_{22}\,x_{32} \\
& - 8x_{22}\,x_{34} - 16x_{22}\,x_{36} - 8x_{22}\,x_{37} - 8x_{22}\,x_{40} - 8x_{22}\,x_{44} \\
& - 16x_{22}\,x_{65} - 8x_{22}\,x_{66} - 16x_{24}^2 - 16x_{24}\,x_{27} - 16x_{24}\,x_{29} \\
& - 32x_{24}\,x_{30} - 16x_{24}\,x_{32} - 16x_{24}\,x_{34} - 32x_{24}\,x_{36} - 16x_{24}\,x_{37} \\
& - 16x_{24}\,x_{40} - 16x_{24}\,x_{44} - 32x_{24}\,x_{65} - 16x_{24}\,x_{66} - 4x_{27}^2 \\
& - 8x_{27}\,x_{29} - 16x_{27}\,x_{30} - 8x_{27}\,x_{32} - 8x_{27}\,x_{34} - 16x_{27}\,x_{36} \\
& - 8x_{27}\,x_{37} - 8x_{27}\,x_{40} - 8x_{27}\,x_{44} - 16x_{27}\,x_{65} - 8x_{27}\,x_{66} \\
& - 4x_{29}^2 - 16x_{29}\,x_{30} - 8x_{29}\,x_{32} - 8x_{29}\,x_{34} - 16x_{29}\,x_{36} \\
& - 8x_{29}\,x_{37} - 8x_{29}\,x_{40} - 8x_{29}\,x_{44} - 16x_{29}\,x_{65} - 8x_{29}\,x_{66} \\
& - 16x_{30}^2 - 16x_{30}\,x_{32} - 16x_{30}\,x_{34} - 32x_{30}\,x_{36} - 16x_{30}\,x_{37} \\
& - 16x_{30}\,x_{40} - 16x_{30}\,x_{44} - 32x_{30}\,x_{65} - 16x_{30}\,x_{66} - 4x_{32}^2 \\
& - 8x_{32}\,x_{34} - 16x_{32}\,x_{36} - 8x_{32}\,x_{37} - 8x_{32}\,x_{40} - 8x_{32}\,x_{44} \\
& - 16x_{32}\,x_{65} - 8x_{32}\,x_{66} - 4x_{34}^2 - 16x_{34}\,x_{36} - 8x_{34}\,x_{37} \\
& - 8x_{34}\,x_{40} - 8x_{34}\,x_{44} - 16x_{34}\,x_{65} - 8x_{34}\,x_{66} - 16x_{36}^2 \\
& - 16x_{36}\,x_{37} - 16x_{36}\,x_{40} - 16x_{36}\,x_{44} - 32x_{36}\,x_{65} - 16x_{36}\,x_{66} \\
& - 4x_{37}^2 - 8x_{37}\,x_{40} - 8x_{37}\,x_{44} - 16x_{37}\,x_{65} - 8x_{37}\,x_{66} \\
& - 4x_{40}^2 - 8x_{40}\,x_{44} - 16x_{40}\,x_{65} - 8x_{40}\,x_{66} - 4x_{44}^2 \\
& - 16x_{44}\,x_{65} - 8x_{44}\,x_{66} - 16x_{65}^2 - 16x_{65}\,x_{66} - 4x_{66}^2 \\
& - x_{148}^2 - x_{157}^2 + 2x_{163} - x_{163}^2 - x_{169}^2 \\
& - x_{173}^2 - x_{177}^2 - x_{181}^2 - x_{185}^2 - x_{189}^2 \\
& - x_{193}^2 - x_{197}^2 - x_{201}^2 - x_{205}^2 + x_{209}^2 \\
& - x_{231}^2 - x_{232}^2 - x_{233} - x_{233}^2
\end{aligned}
$$

$$
\begin{aligned}
R_{173} = \; & -1 - 4x_{106}^2 - 4x_{111}^2 - 4x_{114}^2 - 4x_{117}^2 \\
& - 4x_{120}^2 - 4x_{124}^2 - 4x_{127}^2 + x_{129}^2 - 4x_{131}^2 \\
& - 4x_{135}^2 - 4x_{138}^2 - 4x_{141}^2 - x_{234}^2 - x_{235}^2 \\
& - x_{236} - x_{236}^2
\end{aligned}
$$

B Companion specifications and archive layout

The other two equations are defined completely by their ordered variable and residual lists:

$$P_{239}^{\text{sparse}} = \sum_{i=0}^{174} (S_i)^2, \qquad P_{239}^{\text{low}} = \sum_{i=0}^{174} (T_i)^2.$$

Their respective directories are `equations/P239_sparse/` and `equations/P239_low_height/`. Each contains the entire ordinary polynomial as residuals and as collected coefficients, its exact metrics, and

an explicit restoration certificate. The supplied evaluator distinguishes the profiles by these names.

Path	Contents
`report/report.tex`, `report/report.pdf`	Standalone article and compiled PDF.
`equations/`	Three complete polynomial specifications, all expanded coefficients, and exact final certificates.
`source/`	Changed raw source, Horn factoring, monic projection and translation records, and the intermediate polynomial.
`code/`	Deterministic source producer, frozen endpoint replay, separately written finite checkers, corruption controls, and arithmetic diagnostics.
`proofs/`, `review/`	Unbounded proofs, independent mathematical reviews and actual verification receipts with their precise scopes.
`inherited/factorial_free/`	Attributed immediate predecessor, including ordinary proof closure, logical interfaces, source code, data, and original notices.
`comparisons/`	Literal predecessor polynomials used for the measured comparisons.
`SOURCE_LEDGER.json`, `MANIFEST.sha256.json`	Repository and archive pins, dependency boundary, and file identities.